THIOPENTAL

Extravasation → necrosis

Intra-arterial = gangrene

CHEMICAL NAME Sodium 5-ethyl-5-(pentan-2-yl)-2-sulfanylidene-1,3-diazinane-4,6-dione (thiopental sodium)

CLASS Ultra-short-acting barbiturate IV anaesthetic

TRADE NAMES Pentothal · Thiopentone sodium · Intraval

THERAPEUTIC USES / INDICATIONS

- Induction of general anaesthesia (rapid sequence induction).
- Status epilepticus refractory to other drugs.
- Reduction of raised intracranial pressure (barbiturate coma).

ADMINISTRATION

- **IV only** as freshly prepared 2.5% solution (25 mg/mL).
- Strongly alkaline (pH ~11): ensure IV cannula is in vein — extravasation causes necrosis; intra-arterial injection causes severe pain and gangrene.
- Do not mix with acidic drugs (e.g. suxamethonium) — precipitates.
- Use solution within 24 h.

DOSAGES

- **Induction (adult):** 3–5 mg/kg over 10–15 s, titrated.
- **Children:** 5–8 mg/kg.
- **Status epilepticus:** 3–5 mg/kg bolus, then 3–5 mg/kg/h infusion (ICU).
- Reduce in elderly, hypovolaemia, cardiac disease.

PHARMACOKINETICS

- **Onset:** 30 seconds (one arm-brain circulation time). **Duration:** 5–10 min (redistribution to muscle and fat).
- Slowly metabolised in liver; accumulates with repeated doses → prolonged recovery.
- Excreted in urine. **t½:** 10–12 h.

PHARMACODYNAMICS

- Enhances **GABA-A** receptor activity (prolongs chloride channel opening) → hypnosis.
- Anticonvulsant; reduces cerebral metabolism and ICP.
- No analgesia (may be anti-analgesic at low doses).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hypotension, tachycardia.
- Respiratory depression, apnoea.
- Coughing, hiccups.

ADVERSE EFFECTS (SERIOUS)

- Laryngospasm, bronchospasm.
- Tissue necrosis if extravasated; gangrene if intra-arterial.
- Anaphylaxis.
- Precipitation of acute porphyria.

CONTRAINDICATIONS & CAUTIONS

- Acute porphyria.
- Airway obstruction where ventilation cannot be secured.
- Hypersensitivity to barbiturates.
- **Caution:** severe hypovolaemia, cardiac disease, asthma, myasthenia gravis.

KEY INTERACTIONS

- **Opioids, benzodiazepines, alcohol:** additive depression.
- **Antihypertensives:** profound hypotension.
- Incompatible in syringe with suxamethonium and many acidic drugs.

NURSING MANAGEMENT

- Check cannula patency; watch site during injection.
- Stop immediately and report pain, blanching or swelling at site.
- Monitor airway, SpO_2, BP and ECG.
- Prolonged drowsiness after repeated doses — monitor recovery.
- Check porphyria history.

PATIENT EDUCATION

- Report any pain in the arm during injection.
- Expect drowsiness for several hours; no driving for 24 h.

KEY POINTS

- Ultra-short barbiturate induction agent.
- Alkaline: extravasation necrosis, intra-arterial gangrene.
- Contraindicated in porphyria.
- Also used in refractory status epilepticus.

SOURCES CHECKED **[85]** Thiopental 500mg Powder for Solution for Injection (UK SmPC)

Full references p. 432–457